READORA OPEN ACADEMIC TEXTBOOK SERIES

Deep Learning & Big Data Systems

Neural Architectures, Transformers & Distributed Data Pipelines

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Preface & Curriculum Overview

This textbook has been compiled for undergraduate and postgraduate students enrolled in the **AI & Machine Learning (CSM)** program. It provides structured, rigorous theoretical models, practical case demonstrations, analytical formulas, and chapter-end review exercises. Students are encouraged to practice the core derivations and apply algorithmic workflows to real-world engineering challenges.

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Chapter 1: Deep Neural Network Architectures

1.1 Backpropagation & Automatic Differentiation

Reverse-mode automatic differentiation utilizes the chain rule of calculus to compute exact partial gradients through arbitrary computational graphs.

> ***Key Concept:*** *Non-saturating activation functions such as ReLU mitigate the vanishing gradient dilemma.*

Review Questions & Laboratory Exercises

- Derive the backpropagation equations for a 3-layer fully connected network.
- Compare Batch Normalization with Layer Normalization.

Chapter 2: Convolutional & Sequential Modeling

2.1 CNNs for Spatial Feature Extraction

Convolutional operators enforce translation invariance and spatial locality through parameter sharing and pooling hierarchies.

> ***Key Concept:*** *Self-attention mechanisms replace recurrent loops by calculating scaled dot-product attention matrices across sequence tokens.*

Review Questions & Laboratory Exercises

- Compute the receptive field of a 5-layer CNN architecture.
- Explain multi-head self-attention time complexity.

Chapter 3: Big Data Engineering & Distributed Training

3.1 Data Parallelism & Ring AllReduce

High-throughput model training synchronizes model gradients across compute clusters using ring-based communication topologies.

> ***Key Concept:*** *Distributed file storage ensures fault-tolerant partitioned datasets across commodity worker nodes.*

Review Questions & Laboratory Exercises

- Outline parameter server architecture vs all-reduce data parallelism.
- Explain memory management during gradient accumulation.